

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1511

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 50

Code of Conduct

I S. Amulya / 192321044 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 Hypervisors for a Cloud Datacenter Hosting Mission-Critical Applications (10 Marks)

Introduction

A hypervisor is virtualization software that enables multiple Virtual Machines (VMs) to run on a single physical server. Hypervisors are classified into **Type-1 (Bare-Metal)** and **Type-2 (Hosted)** based on their architecture.

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Installation	Runs directly on hardware	Runs on top of a host operating system
Performance	Very high	Lower due to host OS overhead
Security	More secure	Less secure because host OS can be attacked
Resource Usage	Efficient	Consumes extra resources
Examples	VMware ESXi, Microsoft Hyper-V, Xen	Oracle VirtualBox, VMware Workstation

Performance Analysis

- Type-1 hypervisors communicate directly with hardware, resulting in lower latency and higher throughput.
- Type-2 hypervisors depend on the host operating system, increasing CPU and memory overhead.
- Mission-critical applications require maximum uptime and fast response times, making Type-1 more suitable.

Security Analysis

- Type-1 has a smaller attack surface because there is no general-purpose operating system between hardware and VMs.
- Type-2 inherits vulnerabilities from the host operating system, increasing security risks.
- Type-1 provides stronger VM isolation and secure resource allocation.

Manageability Analysis

- Type-1 supports centralized management, live migration, snapshots, clustering, and automatic failover.
- Type-2 is mainly intended for software testing and desktop virtualization rather than enterprise cloud environments.

Justification

For a cloud datacenter hosting banking, healthcare, or government applications, **Type-1 hypervisors are the better choice** because they provide:

- Better performance
- Stronger security
- Higher availability

- Easier centralized management
- Enterprise scalability

Conclusion

Type-1 hypervisors are ideal for mission-critical cloud infrastructures because they deliver high performance, strong isolation, and enterprise-grade reliability, whereas Type-2 hypervisors are more appropriate for development and testing.

2. Workload Isolation Using Virtualization Architecture and Attack Surfaces

Introduction

Cloud providers often host multiple organizations on the same physical servers. Virtualization enables multiple isolated VMs to securely share hardware resources.

Virtualization-Based Isolation

1. Hypervisor Isolation

Each VM operates independently with its own operating system.

2. Resource Isolation

CPU, RAM, storage, and network resources are allocated separately to each VM.

3. Virtual Network Isolation

Different departments such as Finance, Healthcare, and Student Portals are placed on separate VLANs or virtual networks.

4. Storage Isolation

Each workload uses separate virtual disks and storage permissions to prevent unauthorized access.

Example

- Finance VM → Payroll and financial records
- Healthcare VM → Patient medical records
- Student Portal VM → Student information

Even though all run on the same server, virtualization ensures that one VM cannot directly access another VM's data.

Major Attack Surfaces

1. Hypervisor Attack

- Attackers exploit vulnerabilities in the hypervisor.
- Can compromise multiple VMs simultaneously.
- Prevented by regular patching and secure hypervisor configuration.

2. VM Escape Attack

- Malicious software inside one VM escapes into the hypervisor.
- May gain access to neighboring VMs.
- Prevented through strict isolation and updated virtualization software.

Security Measures

- Role-Based Access Control (RBAC)
- Multi-Factor Authentication
- Encryption of virtual disks
- Network segmentation
- Continuous monitoring

Conclusion

Virtualization architecture enables secure workload isolation while efficiently utilizing hardware resources. Protecting the hypervisor and preventing VM escape attacks are essential for maintaining cloud security.

3. VM Host Utilization Analysis and Corrective Actions (10 Marks)

Given Snapshot

Resource	Value
CPU Utilization	92%
Memory Utilization	88%
Storage I/O Latency	35 ms
Average VM Boot Time	170 s

Analysis

CPU Utilization (92%)

- Extremely high CPU usage
- Indicates CPU contention among multiple VMs
- Applications may experience slow response times.

Memory Utilization (88%)

- Very high RAM consumption
- Hypervisor may begin swapping memory to disk
- Leads to slower VM performance.

Storage I/O Latency (35 ms)

- Storage delay is much higher than the recommended (<10 ms)
- Indicates storage bottlenecks
- Causes slow file access and database operations.

VM Boot Time (170 s)

- Normal VM boot time is usually below one minute.
- Long boot times indicate overloaded CPU, memory, or storage.

Likely Bottlenecks

- CPU overcommitment
- Insufficient RAM
- Slow storage subsystem

Corrective Actions

CPU Optimization

- Migrate some VMs to another host.
- Increase CPU cores.
- Enable load balancing.

Memory Optimization

- Upgrade RAM.
- Enable Dynamic Memory allocation.
- Remove idle VMs.

Storage Improvement

- Replace HDD with SSD or NVMe storage.
- Use faster SAN storage.
- Optimize storage caching.

VM Optimization

- Remove unused snapshots.
- Allocate appropriate VM resources.
- Use thin provisioning carefully.

Conclusion

The system suffers from CPU, memory, and storage bottlenecks simultaneously. Resource balancing, hardware upgrades, and VM optimization will significantly improve performance and reduce boot times.

4. Software Defined Datacenter (SDDC) vs Traditional Datacenter (10 Marks)

Introduction

A **Software Defined Datacenter (SDDC)** virtualizes compute, storage, and networking resources, allowing them to be managed entirely through software.

Traditional Datacenter

- Hardware configured manually
- Fixed networking
- Dedicated storage devices
- Slow provisioning
- Limited scalability

Software Defined Datacenter

Compute Virtualization

- Physical servers host multiple VMs.
- Resources are dynamically allocated.
- Live migration improves availability.

Storage Virtualization

- Multiple storage devices appear as one logical storage pool.
- Automatic storage allocation.
- Better utilization and disaster recovery.

Network Virtualization

- Virtual switches and Software Defined Networking (SDN).
- Rapid network configuration.
- Network policies managed centrally.

Agility Improvements

Traditional	Software Defined Datacenter
Manual provisioning	Automated provisioning
Weeks to deploy servers	Minutes to deploy VMs
Hardware-dependent	Software-controlled
Difficult scaling	Automatic scaling
Manual management	Centralized management

Benefits

- Faster deployment
- Better resource utilization
- Lower operational cost
- High availability
- Simplified disaster recovery
- Easier automation

Conclusion

SDDC greatly improves agility by virtualizing compute, storage, and networking. Organizations can rapidly provision resources, automate operations, and scale services efficiently compared to traditional datacenters.

5. Multi-Cloud Identity Federation and Secure Privacy Design (10 Marks)

Introduction

Multi-cloud environments use services from multiple cloud providers. Identity mismatches occur when users have different credentials across providers, making authentication and authorization difficult.

Problem Analysis

Identity Mismatch

- Different username formats
- Different authentication protocols
- Separate user databases
- Inconsistent access policies

Consequences

- Login failures
- Duplicate accounts
- Poor user experience
- Increased security risks
- Administrative complexity

Federation-Based Solution

Central Identity Provider (IdP)

Use a single trusted Identity Provider to authenticate users.

Single Sign-On (SSO)

Users log in once and access multiple cloud providers without repeated authentication.

Federation Standards

- **SAML (Security Assertion Markup Language):** Secure exchange of authentication information.
- **OAuth 2.0:** Secure delegated authorization for applications.
- **OpenID Connect (OIDC):** Authentication layer built on OAuth 2.0.

Privacy Design

- Encrypt identity data during transmission and storage.
- Use Multi-Factor Authentication (MFA).
- Implement Role-Based Access Control (RBAC).
- Share only the minimum required user attributes (principle of least privilege).
- Maintain audit logs and continuous monitoring.

Secure Identity Architecture

User



Identity Provider (IdP)



SSO Authentication



Cloud Provider A

Cloud Provider B

Cloud Provider C

Benefits

- One secure login
- Consistent identity across providers
- Improved security
- Reduced administrative effort
- Better privacy protection
- Simplified access management

Conclusion

Identity federation using **SAML**, **OAuth 2.0**, and **OpenID Connect**, combined with SSO, MFA, RBAC, encryption, and audit logging, provides a secure and scalable solution for resolving identity mismatches in multi-cloud environments while protecting user privacy.